

SOFTWARE REQUIREMENTS SPECIFICATION (SRS)

SENTRA

Intelligent Monitoring & Protection System

Project Proposal / Partial SRS

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) document defines the requirements, functionality, hardware, software, system behavior, limitations, and expected operation of SENTRA – Intelligent Monitoring & Protection System.

SENTRA is an AI-assisted surveillance system designed to monitor an area around a home where motorcycles, bikes, and cars are parked.

The system uses a camera to capture the monitored area and Artificial Intelligence to detect people and vehicles. It identifies predefined events, records important events, stores relevant information, and can notify the owner when a configured event occurs.

1.2 Project Purpose

The purpose of SENTRA is to reduce the need for continuous manual CCTV monitoring.

Traditional CCTV systems mainly record footage, while the owner has to watch the camera or manually search recorded footage when an incident occurs.

SENTRA adds an AI-based detection layer so that important events can be identified automatically and presented to the owner.

2. Project Scope

2.1 In Scope

The initial version of SENTRA will include:

- Camera-based monitoring.
- Live video monitoring.
- Person detection.
- Motorcycle/bike detection.
- Car detection.
- Predefined event detection.
- Event recording.
- Local storage.
- Event history.
- Time-based event information.
- Notifications for configured events.
- Web/mobile monitoring interface.
- AI processing using a laptop/PC during the initial prototype.

2.2 Out of Scope

The following features are not required for the initial version:

- Guaranteed theft detection.
- Guaranteed identification of criminal activity.

Face recognition.
Automatic identification of the person.
Police/emergency service integration.
Advanced multi-camera surveillance.
Unlimited behavior recognition.
Cloud storage as a mandatory requirement.
Fully standalone hardware in the first prototype.

These features may be considered in future versions if required.

3. Problem Description

3.1 Existing Problem

Homeowners may park motorcycles, bikes, and cars outside or near their homes.

A traditional CCTV camera can record this area, but it normally does not automatically tell the owner that an important event has happened.

The owner may therefore need to:

- Continuously watch the camera.
- Check the footage manually.
- Search through long recordings.
- Identify important events by themselves.

This process is time-consuming and important events may be missed.

4. Proposed Solution

SENTRA will monitor the selected area through a camera.

The camera feed will be processed by an AI system.

The AI will detect relevant objects such as:

- Person
- Motorcycle/Bike
- Car

The system will use these detections to identify predefined events.

When a configured event occurs, SENTRA will:

1. Detect the event.
2. Record relevant information.
3. Store the event.
4. Generate an alert when required.
5. Display the event in the monitoring interface.

5. System Overview

The basic system flow is:

Camera → AI Processing → Object Detection → Event Detection → Recording/Storage → Notification → Web/
Mobile App

Initial Prototype

Camera → Laptop/PC → AI → Backend → Web/Mobile App

The laptop/PC will initially perform AI processing because this reduces the initial hardware cost and makes development and testing easier.

Later, the processing system can potentially be moved to an embedded device such as a Raspberry Pi or NVIDIA Jetson.

6. Users of the System

6.1 System Owner

The primary user will be the owner of the monitored property.

The owner can:

- View the camera.
- View detected events.
- Review event history.
- Receive notifications.
- Check recorded events.
- Monitor the status of the system.

7. Functional Requirements

Functional requirements describe what SENTRA must do.

FR-01: Camera Connection

The system shall connect to a supported camera.

The camera shall provide video input to the monitoring system.

FR-02: Live Video

The system shall provide access to the live camera feed through the monitoring interface.

The owner shall be able to view the monitored area.

FR-03: Person Detection

The AI system shall detect people appearing within the camera's monitored area.

The system shall generate a detection result when a person is identified.

FR-04: Motorcycle/Bike Detection

The AI system shall detect motorcycles/bikes appearing within the monitored area.

The detection shall be associated with the relevant camera frame/event.

FR-05: Car Detection

The AI system shall detect cars appearing within the monitored area.

The detection shall be associated with the relevant camera frame/event.

FR-06: Event Detection

The system shall convert relevant detections into predefined events.

Examples may include:

- Person approaching a monitored vehicle.
- Person remaining near a monitored vehicle.
- Vehicle entering the monitored area.
- Vehicle leaving the monitored area.

The exact events shall be defined during implementation and testing.

FR-07: Event Timestamp

Every stored event shall contain a date and time.

Example:

Person detected near motorcycle – 10:32 PM

FR-08: Event Recording

When a configured event occurs, the system shall save relevant recording or selected video evidence.

The purpose is to allow the owner to review the event later.

FR-09: Event History

The system shall maintain a history of detected events.

The event history may contain:

- Event type.
- Date.
- Time.
- Detected object.
- Relevant recording/clip.
- Event status.

FR-10: Notifications

The system shall generate a notification when a configured event occurs.

Example:

Alert: Person detected near motorcycle.

The notification mechanism will depend on the selected implementation.

FR-11: Monitoring Interface

The system shall provide a web or mobile interface.

The interface shall allow the owner to access important monitoring functions.

FR-12: Event Review

The owner shall be able to select an event from the event history and review its available information or recording.

FR-13: Local Storage

The system shall support local storage for selected recordings and event information.

Storage may be provided through:

- Computer storage.
- SSD.
- microSD card where supported by the selected hardware.

FR-14: System Status

The interface should provide basic system status information, such as:

- Camera connected/disconnected.
- AI processing status.
- Monitoring status.
- Storage availability where supported.

8. Event Detection Logic

SENTRA will not claim that it can perfectly determine whether a theft has occurred.

Instead, the system will detect predefined suspicious or important events based on available visual evidence.

For example:

Normal Situation

Person → walks past vehicle → leaves

The system may simply record the detection.

Configured Event

Person → approaches motorcycle → remains near motorcycle

The system can create a configured event and send an alert.

This approach keeps the system realistic and avoids claiming that AI can guarantee that a theft is taking place.

9. Recording Requirements

9.1 Recording

The system shall support recording of relevant camera footage.

9.2 Event-Based Recording

The system should prioritize selected events instead of requiring the owner to manually search through all footage.

9.3 Storage

Recorded information shall be stored using the selected local storage system.

9.4 Storage Management

The system should provide a mechanism for managing stored recordings when storage becomes limited.

The exact retention period will be determined during implementation according to available storage capacity.

10. AI Requirements

10.1 Object Detection

The AI component shall detect the required object categories:

Person

Motorcycle/Bike
Car

10.2 Detection Result

Each detection should provide information required by the event-detection component.

Possible information includes:

- Object type.
- Detection time.
- Object location within the frame.
- Detection confidence.

10.3 Event Generation

The system shall use detection results to determine whether a predefined event has occurred.

11. Backend Requirements

The backend will act as the communication layer between the AI processing system and the web/mobile interface.

The backend shall:

- Receive relevant event information.
- Store event information.
- Provide event information to the application.
- Handle communication between system components.
- Support notification processing where required.

12. Database Requirements

The system shall maintain structured information about detected events.

A basic event record may contain:

Field	Description
Event ID	Unique identifier
Event Type	Type of detected event
Object Type	Person, bike, or car
Date	Date of event
Time	Time of event
Recording Path	Location of related recording
Status	Event/alert status

13. User Interface Requirements

The interface should be simple and easy to understand.

13.1 Dashboard

The dashboard should provide access to:

- Live camera.
- Current monitoring status.
- Recent events.
- Alerts.
- Event history.

13.2 Live Monitoring

The owner should be able to view the current camera feed.

13.3 Event History

The owner should be able to see previous events.

13.4 Alerts

Important alerts should be clearly displayed.

14. Notification Requirements

The system shall support notifications for configured events.

A notification should provide useful basic information.

SENTRA Alert
Event: Person detected near motorcycle
Time: 10:32 PM

The exact notification method will depend on the final implementation.

15. Hardware Requirements

15.1 Camera

A 1080p USB camera or suitable camera module can be used.

Purpose: Capture live video.

15.2 Laptop/PC

A laptop or PC will be used during the initial prototype.

Purpose:

- AI processing.
- Development.
- Backend.
- Testing.
- Local storage where required.

15.3 Storage

Storage may include:

- Internal computer storage.
- SSD.

- microSD card when supported by the selected camera/embedded hardware.

15.4 Network

Wi-Fi or another network connection will be required for communication between the system components and the application.

15.5 Future Hardware

A Raspberry Pi or NVIDIA Jetson may be used in a future version for standalone edge processing.

16. Software Requirements

The system will require software components for:

- Camera/video processing.
- Computer vision.
- AI object detection.
- Event detection.
- Backend/API.
- Database/event logging.
- Web/mobile interface.
- Notification handling.

The exact programming languages, frameworks, AI model, and database technology will be selected during the implementation phase.

17. Non-Functional Requirements

17.1 Performance

The system should process the camera feed with reasonable delay for real-time monitoring.

17.2 Reliability

The system should continue monitoring when the camera and required system components are available.

17.3 Usability

The interface should be simple enough for a normal user to understand without technical knowledge.

17.4 Scalability

The architecture should allow additional event types or cameras to be considered in future versions.

17.5 Maintainability

The AI detection, backend, storage, and interface components should be separated so that individual components can be updated without unnecessarily changing the complete system.

17.6 Security

Access to the monitoring interface and stored information should be protected from unauthorized users.

18. System Workflow

Step 1 – Camera Starts

The camera starts capturing the monitored area.

Step 2 – Video Processing

The camera feed is sent to the AI processing system.

Step 3 – Object Detection

The AI analyzes the video and detects people, bikes, and cars.

Step 4 – Event Analysis

The system checks whether the detected activity matches any predefined event.

Step 5 – Event Creation

If a relevant event occurs, the system creates an event record.

Step 6 – Recording

Relevant footage or evidence is stored.

Step 7 – Notification

If the event requires an alert, a notification is generated.

Step 8 – User Review

The owner can open the application and review the event and available recording.

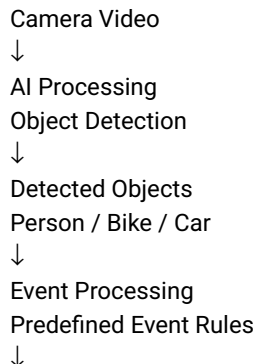
19. Example Scenario

Scenario: Person Approaches a Parked Motorcycle

1. The camera observes the motorcycle.
2. A person enters the camera view.
3. AI detects the person.
4. AI detects the motorcycle.
5. The system analyzes their positions and movement.
6. The predefined event condition is satisfied.
7. SENTRA creates an event.
8. The relevant information is stored.
9. A notification is generated.
10. The owner can review the event through the application.

20. Data Flow

Input



```

graph TD
    A[Output  
Event + Timestamp + Recording] --> B[Storage  
Local Storage / Database]
    B --> C[User  
Web/Mobile Interface + Notification]
  
```

21. Storage Requirements

SENTRA will primarily support local storage for the initial version.

Possible storage locations include:

- PC/laptop storage.
- SSD.
- microSD card with compatible hardware.

The system should avoid unnecessary permanent storage of every piece of video if event-based recording is being used.

The exact recording duration and storage retention policy will be finalized during implementation according to available storage.

22. Error Handling

The system should handle common failures.

Camera Disconnected

If the camera becomes unavailable, the system should indicate that the camera is disconnected.

Network Failure

If network communication fails, the system should indicate the communication problem and recover when the connection becomes available.

Storage Full

If available storage becomes insufficient, the system should notify the user or apply the selected storage-management policy.

AI Processing Failure

If AI processing stops, the system should indicate that monitoring is not functioning normally.

23. Privacy Considerations

Because SENTRA processes camera footage around a home, privacy should be considered.

The system should:

- Restrict access to authorized users.
- Protect stored recordings.
- Avoid unnecessary collection of personal information.
- Clearly define where recordings are stored.

Use recordings only for the intended monitoring purpose.

24. Limitations

SENTRA will have some limitations.

24.1 AI Is Not Perfect

AI detection may produce incorrect or missed detections.

24.2 Environmental Conditions

Performance may change because of:

- Darkness.
- Poor lighting.
- Rain.
- Camera obstruction.
- Objects blocking the view.
- Poor camera positioning.

24.3 No Guaranteed Theft Detection

The system detects predefined visual events. It cannot guarantee that a theft or crime is occurring.

24.4 Camera Coverage

The system can only monitor areas visible to the camera.

24.5 Storage Capacity

The amount of recording that can be retained depends on available storage.

25. Testing Requirements

25.1 Person Detection Test

Test whether the system detects a person entering the monitored area.

25.2 Bike Detection Test

Test whether the system detects a motorcycle/bike correctly.

25.3 Car Detection Test

Test whether the system detects a car correctly.

25.4 Event Detection Test

Test predefined events involving people and vehicles.

25.5 Recording Test

Verify that relevant events are recorded and stored.

25.6 Notification Test

Verify that configured events generate notifications.

25.7 Storage Test

Verify that recordings and event information are stored correctly.

25.8 Application Test

Verify that the owner can view the live camera and event history.

26. Development Phases

Phase 1 – Camera Setup

Set up the camera and establish video streaming.

Phase 2 – AI Detection

Implement person, bike, and car detection.

Phase 3 – Event Detection

Define and test a limited set of vehicle-related events.

Phase 4 – Recording and Storage

Implement event recording and storage.

Phase 5 – Backend

Connect AI processing with the backend and database.

Phase 6 – Monitoring Interface

Develop the web/mobile interface.

Phase 7 – Notifications

Implement notifications for configured events.

Phase 8 – Testing

Test the complete system in a controlled environment and evaluate detection performance.

Phase 9 – Hardware Upgrade

If required, move AI processing from the laptop/PC to suitable edge hardware.

This follows the same general staged approach as the original proposal: start with a low-cost camera + PC prototype, validate the AI pipeline, and only then consider embedded hardware.

27. Expected Final Product

The final prototype of SENTRA will consist of:

Physical Camera + AI Video Analysis + Event Detection + Recording/Storage + Alert System + Web/Mobile Interface

The owner will be able to:

- View the camera.
- Monitor the parking area.
- See detected people and vehicles.
- Review detected events.
- Access relevant recordings.
- Receive alerts for configured events.

28. Future Enhancements

Possible future enhancements include:

- Multiple camera support.
- Standalone Raspberry Pi/Jetson implementation.

Improved event detection.
Number plate recognition.
Better low-light detection.
Cloud backup.
Voice interaction.
More advanced analytics.
Additional types of monitored areas.

These features are considered future possibilities and are not required for the initial prototype.

29. Success Criteria

SENTRA will be considered successful if the prototype can:

1. Successfully receive video from the camera.
2. Detect people.
3. Detect motorcycles/bikes.
4. Detect cars.
5. Identify the selected predefined events.
6. Record relevant events.
7. Store event information.
8. Display event history.
9. Generate configured alerts.
10. Allow the owner to access the monitoring interface.

30. Final System Concept

The complete concept of SENTRA can be summarized as:

“SENTRA is an AI-assisted monitoring system that watches a home parking area, detects people and vehicles, identifies predefined important events, records relevant evidence, and alerts the owner when necessary.”

The key difference from traditional CCTV is that SENTRA does not only record the scene; it also analyzes the camera feed to help identify important events automatically.